Appendix: A Continual Learning Framework for Uncertainty-Aware Interactive Image Segmentation

In this Appendix, we summarize the notations of variables in Table 2 and describe the inference process in Algorithm 1. We then provide additional discussion of related concepts and methods, and present an ablation study to evaluate some key components of the framework, including uncertainty guided interactions and task-aware kernel activation for initial segmentation. We also provide illustrative examples of uncertainty-aware interactive segmentation based on simulated and real user interactions.

Table 2: Summary of Notations

$\mathcal{T}_t$	task t
X_t	Images in task t
A_t, U_t	ground truth labels of annotated and unannotated pixels in task t
h_k^l	k-th channel of the feature map at layer l
W_k^l	k-th convolutional kernel at layer l
z_k^l	binary mask controlling layer l kernel k's activation
$a_{k,t}^l, b_{k,t}^l$	Beta parameters of v_k^l in task t
$\mu_{ki}^l, (\sigma_{ki}^l)^2$	Gaussian mean and variance of W_k^l's element i in task t
π_k^l	Bernoulli parameter of z_k^l
$\mathbf{r}_m, \hat{\mathbf{r}}_m$	RGB vector and predicted RGB vector of pixel m
$\bar{\boldsymbol{\theta}}_m$	mean predicted probability vector of pixel m
$\mathbf{y}_m$	ground truth one-hot label vector of pixel m
$\boldsymbol{\theta}_m^*$	propagated probability vector of pixel m
$\mathbf{y}_m^*$	propagated one-hot label vector of pixel m

Algorithm 1 Inference Process for Task t

Given parameters $a_{k,t-1}^l$, $b_{k,t-1}^l$, $\pi_{k,t-1}^l$, $\mu_{ki,t-1}^l$, $\sigma_{ki,t-1}^l$;
Given new image X_t;
for layer l=1:L **do**
  for kernel k=1:K **do**
    Sample $v_{k,t}^l$, $z_{k,t}^l$, $W_{k,t}^l$ using (2), (8), (3);
  end for
end for
Predict recovered image and evaluate loss using (6), (**??**);
Update $a_{k,t}^l$, $b_{k,t}^l$, $\mu_{ki,t}^l$, $\sigma_{ki,t}^l$, $\pi_{k,t}^l$;
do
  for layer l=1:L **do**
    for kernel k=1:K **do**
      Sample $v_{k,t}^l$, $z_{k,t}^l$, $W_{k,t}^l$ using (2), (8), (3);
    end for
  end for
  Predict segmentation and uncertainty using (10), (**??**),
  Collect user annotations and propagate using (11);
  Evaluate loss using (6), (**??**) (15);
  Update $a_{k,t}^l$, $b_{k,t}^l$, $\mu_{ki,t}^l$, $\sigma_{ki,t}^l$, $\pi_{k,t}^l$;
while user unsatisfied with segmentation

Detailed Comparison with Related Work

We further differentiate the proposed CLIS framework from some important related concepts, focusing on the problem setting, model training, and performance evaluation. We then highlight the key technical difference between CLIS and some specific methods as compared in the main paper.

- CL-new class: Continual learning with new classes, or incremental class learning, is one scenario of continual learning. It formulates a series of tasks in which some new classes are available to the model. The model should be able to recognize the new classes without forgetting the previous ones.
- CL-new instance: Continual learning with new instances is another scenario of continual learning. All classes are known in pre-training, and each subsequent task may contain instances with a different subset of classes.
- Active learning: An active learning algorithm allows human interaction by querying a user to label new data points to improve the algorithm's performance efficiently. Active learning can be applied to interactive segmentation, and it typically queries a user to provide pixel-wise labels of a whole image. It is different from interactive segmentation, where a user only annotates a few pixels.
- Online learning: An online learning algorithm aims to optimize the predictions over data instances sequentially. It should be noted that online learning also suffers from catastrophic forgetting problems, which is similar to continual learning. However, online learning does not emphasize task-based setups, and keeps learning over time in a streaming fashion.

Generally speaking, the setting of CLIS is similar to CL-new instance, but is customized for interactive segmentation, because it processes data in small batches and actively queries user annotation on selective pixels. The setting of CLIS is also similar to conventional interactive segmentation, but it focuses on updating model parameters to learn from user annotation, rather than post-processing the predictions only.

Table 3 further highlights the key technical difference between the proposed CLIS framework and most related methods. In particular, CLIS uses a Bayesian non-parametric mask to control activated kernels for different tasks and regularizes the parameters using KL divergence to the prior distribution learned from previous tasks. LwF and EWC do not manifest neural resource allocation mechanisms, while PackNet lacks specific regularizations for knowledge preservation. In addition, different from all existing work, CLIS leverages a task-aware design to infer the optimal mask given new images without annotated labels. It further provides uncertainty information to guide user annotations, while the related methods do not contain those components.

Ablation Study

We conduct a systematic ablation study to evaluate some key components of the framework to illustrate their contribution to the performance.

Table 3: Comparison on key technical components

	Neural resource allocation	Regularization	Task-aware design for new data	Uncertainty estimation
CLIS	Kernel activation mask	KL divergence	Yes	Yes
LwF[1]	N/A	Distillation loss	N/A	N/A
EWC[2]	N/A	Weighted panelty for parameters	N/A	N/A
PackNet[3]	Network pruning	N/A	N/A	N/A

Uncertainty Guided Interactions. Our first ablation study analyzes the effectiveness of uncertainty guided interaction. Given uncertainty estimation, we simulate user interactions to annotate the eight erroneous segmented areas with the highest uncertainty, up to two iterations. Without uncertainty estimation, eight erroneous segmented areas are randomly selected and provided the correct labels. We compare the two strategies in terms of basic performance. The results in Table 4 show that uncertainty-guided annotations consistently outperform randomly choosing erroneous segmented areas and providing annotations.

Table 4: Ablation study that compares uncertainty guided interaction with random annotations on the erroneous segmented areas: segmentation accuracy is recorded after two iterations, and averaged over all tasks.

	Dataset 1		Dataset 2		Dataset 3	
	Acc.	IoU	Acc.	IoU	Acc.	IoU
Guided	**86.7**	**77.8**	**91.5**	**85.0**	**82.8**	**72.1**
Random	85.6	75.1	88.3	79.5	81.6	69.7

Task-Aware Kernel Activation. Our second ablation study analyzes the effectiveness of using image information for determining the optimal mask before user interactions. An alternative strategy is plugging in the posterior parameters inferred from the previous task as the prior parameters for the current task to determine the mask. This strategy has a drawback as the posterior parameters may not be suitable for the current task. As is shown in Table 5, this strategy under-performs the proposed framework in terms of forward knowledge transfer, which is evaluated using images from future tasks without user interactions.

Table 5: Ablation Study that compares forward knowledge transfer performance using task-aware kernel activation with using kernel activation inferred from previous task

	Dataset 1		Dataset 2		Dataset 3	
	Acc.	IoU	Acc.	IoU	Acc.	IoU
Task-aware	**79.2**	**65.5**	**84.5**	**75.0**	**77.8**	**63.4**
Previous Task	78.7	64.2	81.8	72.8	76.7	62.3

Additional Details of Experiments

Overconfidence usually results from the out-of-distribution (OOD) issue. The uncertainty estimation methods that we used can detect OOD. In experiments, we do not observe significant issues of overconfidence. However, it should be noted that an uncertainty estimation method may not accurately detect all OOD cases. Practically, one solution is to use ensemble methods (i.e., use multiple uncertainty estimation methods jointly, because the chance of missing an OOD case by all methods is lower than relying on a single method). Another solution is to allow end-users to make final judgments.

Illustration of Simulated and Real-User Interaction

We provide illustrative examples of the simulated user interactions in Figure 5. Given new images, the model generates initial segmentation (column 3). Then the simulated users annotate on selective areas (column 5) guided by uncertainty information (column 4). After that, the model refines segmentation results (column 6) based on user annotation.

In addition, we provide a simple user study for qualitative illustration. As shown in Figure 6, the user interface includes a pop-up window visualizing the uncertainty map, which allows users to click for annotation, and a textbox for typing in the corresponding labels. Then the information of annotations is passed back to the model for refinement. Given the images from Figure 5, we visualize the clicks collected from real-users and the refined segmentation maps in Figure 7. In general, the refined results are similar to those from simulated annotations.

However, it should be noted that real-user interaction may be affected by noisy factors (e.g., humans usually pay more attention to the center of images and objects with high-contrast), making it not ideal for quantitative comparison. In contrast, the simulated interaction is suitable for quantitative evaluation by providing a controlled environment and eliminating the factors that could affect human actions.

Reference

[1] Li, Z., & Hoiem, D. (2017). Learning without forgetting. IEEE transactions on pattern analysis and machine intelligence, 40(12), 2935-2947.

[2] Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., & Hassabis, D. (2017). Overcoming catastrophic forgetting in neural networks. Proceedings of the national academy of sciences, 114(13), 3521-3526.

[3] Mallya, A., & Lazebnik, S. (2018). Packnet: Adding multiple tasks to a single network by iterative pruning. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (pp. 7765-7773).

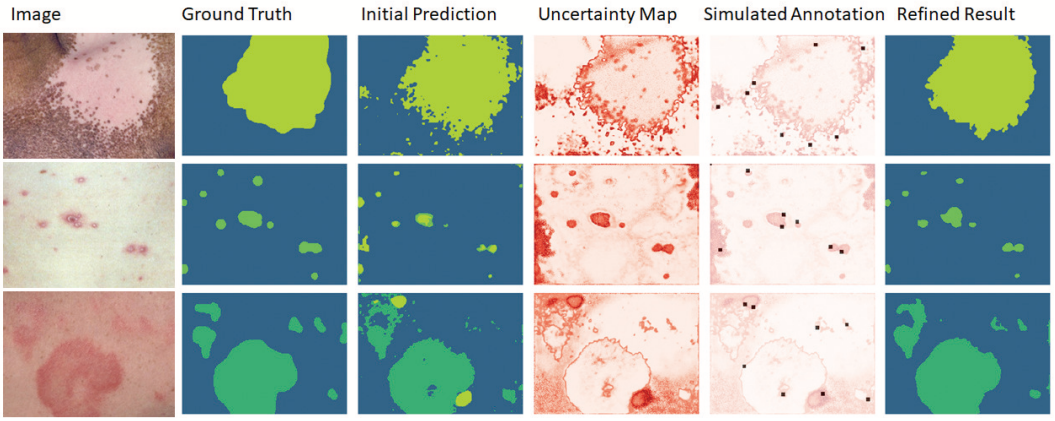

Figure 5: Illustrative examples of uncertainty-aware interactive segmentation: Different semantic classes are denoted by different colors in segmentation maps (columns 3 and 6). Areas with high uncertainty are highlighted in red (column 4), and annotated areas are highlighted in black (column 5).

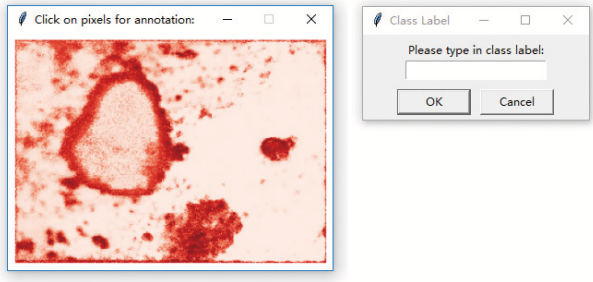

Figure 6: An illustrative example of user interface: users click on the pop-up window and provide the corresponding pixel label on the textbox.

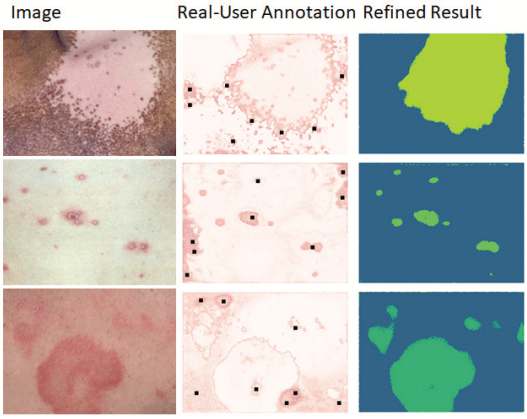

Figure 7: Illustrative examples of interactive segmentation with real-users: The imaged and ground-truth labels are the same as those in Figure 5. Areas with high uncertainty are highlighted in red, and annotated areas by users are highlighted in black (column 2).